CAPSTONE PROJECT

WEBSITE – DEMO BLAZE

Link: https://www.demoblaze.com

ABOUT

Demo Blaze is a sample e-commerce web application used for practicing and demonstrating web automation testing skills. It simulates an online shopping platform where users can browse products, add items to cart, navigate categories, place orders, and interact with contact functionality.

The website is widely used for automation practice because it contains real-world workflows such as authentication, cart management, checkout, session handling, and product navigation.

OBJECTIVE

This project aims to automate and test the Demo Blaze website using the Playwright automation framework.

The main goal is to validate critical e-commerce workflows including authentication, product browsing, cart management, checkout process, navigation flow, search functionality, session handling, and contact features.

Automation helps reduce manual effort, improve testing speed, increase accuracy, and improve application reliability.

TECH STACK

Category	Technology
Automation Framework	Playwright
Programming Language	JavaScript
Runtime Environment	Node.js
IDE	VS Code

Category	Technology
Package Manager	npm
Browser Support	Chromium, Firefox, WebKit
Reporting Tool	Allure Report
CI/CD Integration	GitHub Actions
Containerization	Docker

CORE FUNCTIONALITIES

1. User authentication and session management
2. Product listing and product details validation
3. Product category navigation
4. Product search functionality
5. Add to cart and remove from cart functionality
6. Cart validation and quantity management
7. Checkout workflow and order placement
8. Contact form interaction and navigation handling

SERVICES

1. Authentication Service
2. Product Service
3. Order Service
4. Cart Management Service
5. Contact Service
6. Navigation Service
7. Checkout Service
8. User Session Management Service

SCOPE OF TESTING

The scope of this capstone project is limited to frontend functionality of the Demo Blaze website.

The focus is functional UI automation using Playwright.

It excludes backend validation, database testing, performance testing, and security testing.

IN SCOPE

- Authentication workflows
- Product listing and product detail validation
- Product category navigation
- Search behavior validation
- Add to cart and remove from cart functionality
- Checkout and order placement flow
- Contact functionality validation
- Session persistence and navigation handling

OUT OF SCOPE

- Real payment processing
- Backend/database validation
- Performance testing
- Load testing
- Security testing
- Server-side validation

TEST CASE PRIORITY

Priority	Description
High Priority (P1)	Critical business flows such as authentication, cart operations, checkout, and session handling
Medium Priority (P2)	Product validation, navigation workflows, search functionality
Low Priority (P3)	UI validations, optional flows, secondary interactions

SERVICE-WISE TEST CASE DISTRIBUTION

No	Service / Module	Testing Coverage	Number of Test Cases
1	Authentication Service	Login validation, signup validation, invalid credentials, logout flow, authentication persistence, session handling	15
2	Product Service	Product listing, product details, image validation, price validation, category validation, product navigation	15
3	Search Service	Search functionality, empty search, invalid search, result validation, reset search, repeated searches	15
4	Cart Management Service	Add/remove products, multiple products, cart badge validation, quantity handling, persistence validation	15
5	Contact Service	Contact form validation, invalid inputs, popup validation, field validation, submission handling	15
6	Navigation Service	Home navigation, category navigation, forward/back navigation, routing validation, URL validation	15

No	Service / Module	Testing Coverage	Number of Test Cases
7	Checkout Service	Place order workflow, form validation, confirmation flow, popup handling, error validation	15
8	User Session Management Service	Session persistence, refresh validation, logout validation, multi-page behavior, state validation	15

TOTAL TEST CASES = 120

TEST DATA STRATEGY

Since Demo Blaze is a public testing website:

- Use dummy credentials and reusable test data
- Avoid personal or financial information
- Use reusable automation datasets
- Maintain separate test data files
- Ensure stable execution across browsers
- Reset application state whenever required

PROJECT GOALS

- Build a scalable Playwright automation framework
- Cover 120 functional test cases across 8 services
- Validate end-to-end e-commerce workflows
- Integrate Allure reporting
- Implement CI/CD using GitHub Actions
- Enable Docker-based execution
- Support execution across Chromium, Firefox, and WebKit

• Improve maintainability and reusability

CONCLUSION

This capstone project focuses on automating the Demo Blaze website using Playwright to validate real-world e-commerce workflows such as authentication, product browsing, navigation, cart management, checkout, and session handling.

The project emphasizes functional testing using a structured automation framework integrated with CI/CD, Allure reporting, and Docker support.

The objective is to demonstrate scalable automation practices and build a maintainable testing architecture suitable for real-world QA environments.

This version should look much cleaner when pasted into Word/PDF because the important sections are structured into tables rather than long lists.